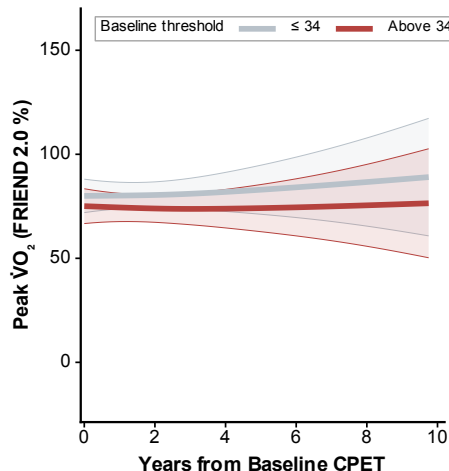
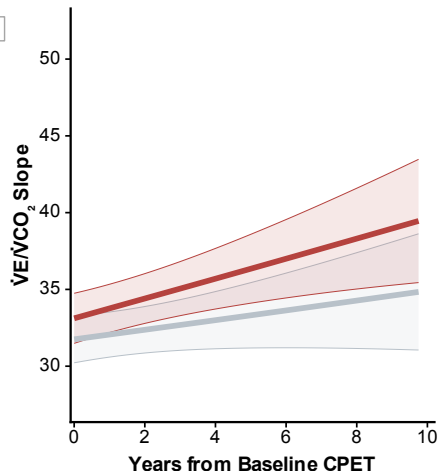


(A) Peak $\dot{V}O_2$ Baseline LAVi ≤ 34 mL/m² versus above 34 mL/m²**(B) $\dot{V}E/\dot{V}CO_2$ slope**Baseline LAVi ≤ 34 mL/m² versus above 34 mL/m²**(C) Longitudinal Model Summary**

Model component	edf	F statistic	P value
Peak $\dot{V}O_2$			
Time smooth	1.24	0.10	0.747
Threshold group	—	2.07	0.151
Time x threshold group	1.00	0.13	0.718
$\dot{V}E/\dot{V}CO_2$ slope			
Time smooth	1.04	2.32	0.124
Threshold group	—	3.14	0.077
Time x threshold group	1.03	1.29	0.256

Supplemental Figure. Unadjusted longitudinal cardiopulmonary trajectories stratified by binary baseline LAVi (mL/m²) status. **(A)** GAMM-derived Peak $\dot{V}O_2$ trajectories. **(B)** GAMM-derived $\dot{V}E/\dot{V}CO_2$ slope trajectories. **(C)** Summary of the key status and interaction terms for both models. Baseline threshold groups were defined as LAVi ≤ 34 mL/m² versus above 34 mL/m². These sensitivity GAMMs were unadjusted for demographic, medication, or diabetes covariates and included only the time smooth, above-threshold effect, time-by-threshold interaction, and subject-level random intercept.